Balmer Series and Rydberg Constant

Aim

To measure the wavelengths of the Balmer Series and calculate Rydberg constant.

Analysis

The diffraction grating disperses the light so that the diffraction angle θ, depends on the wavelength of light λ, and the grating spacing d, according to the grating equation,

$$d \sin \theta = m\lambda$$

The wavelengths of the main lines of mercury are given in the table. We calculate value of d using these values.

Colour	λ in nm	d	σ_d
Violet	404	1921.79	39.43
Blue	436	1932.96	33.40
Cyan	492	1939.98	25.69
Green	546	1945.36	29.76
Yellow	578	1947.31	22.99

Table 1: The values are given before correction

This gives the average value of 1940 ± 12. However this is not the actual value of d. The beam observed from the slit was tilted thus we add a small correction term to get the actual value. It diagram below explains the same.

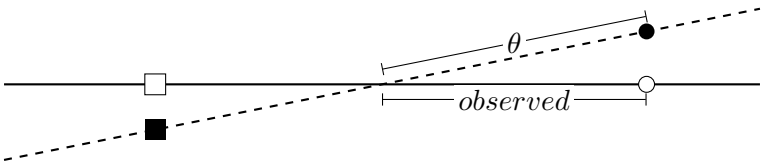


Figure 1: The dotted line has points with the actual angle

We find the correction term using the data from the original dataset. We take ratio of the angle in both reading for the same colour and multiply that ratio across to get the final

value. There might be a shift in angle in here as well but the measurement for the hydrogen lamp was taken in the same settings which gets corrected automatically. We find the ratio in measurements of the datasets is 0.9435 (which gives the angle of deviation as 19.35°). This gives us the corrected version of the table

Color	λ	d	σ_d
Violet	404	2032.54	38.06
Blue	4346	2043.63	31.00
Cyan	492	2049.55	21.44
Green	546	2053.57	24.20
Yellow	578	2054.47	15.91

Now we find the average.

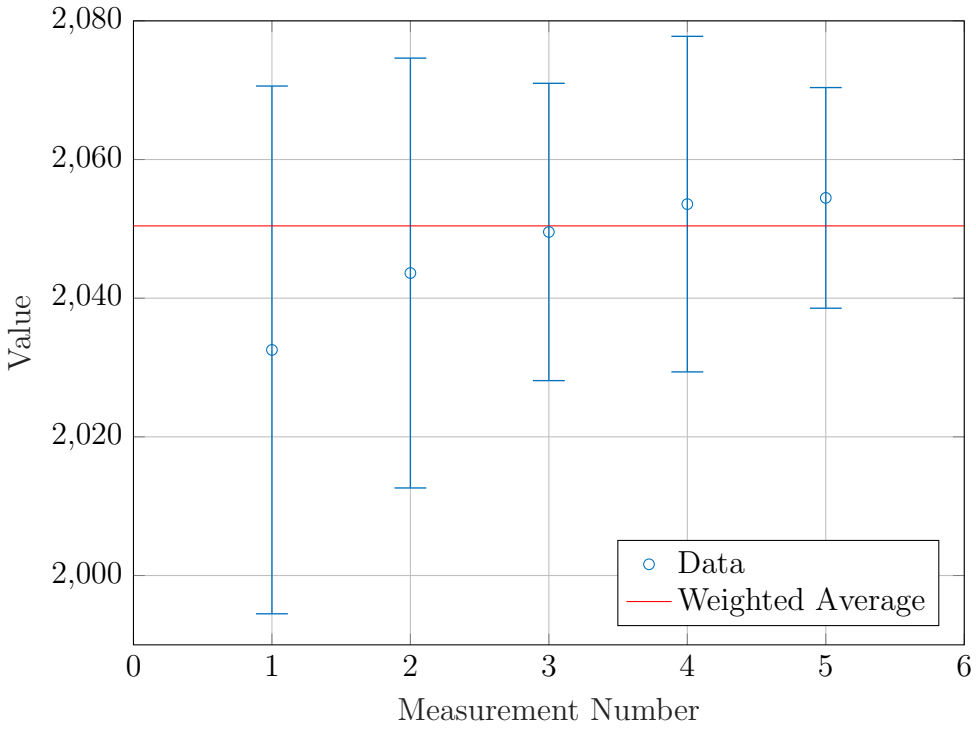


Figure 2: This gives the average value of 2050.42 ± 10.23

We use the same formula to find the Rydberg constant.

Colour	λ (in nm)	$y(1/\lambda)$	$x(0.25 - 1/n_i^2)$
Violet	463.47	2.16E+06	0.2222
Blue	498.25	2.01E+06	0.2100
Green	534.03	1.87E+06	0.1875
Red	674.74	1.48E+06	0.1389

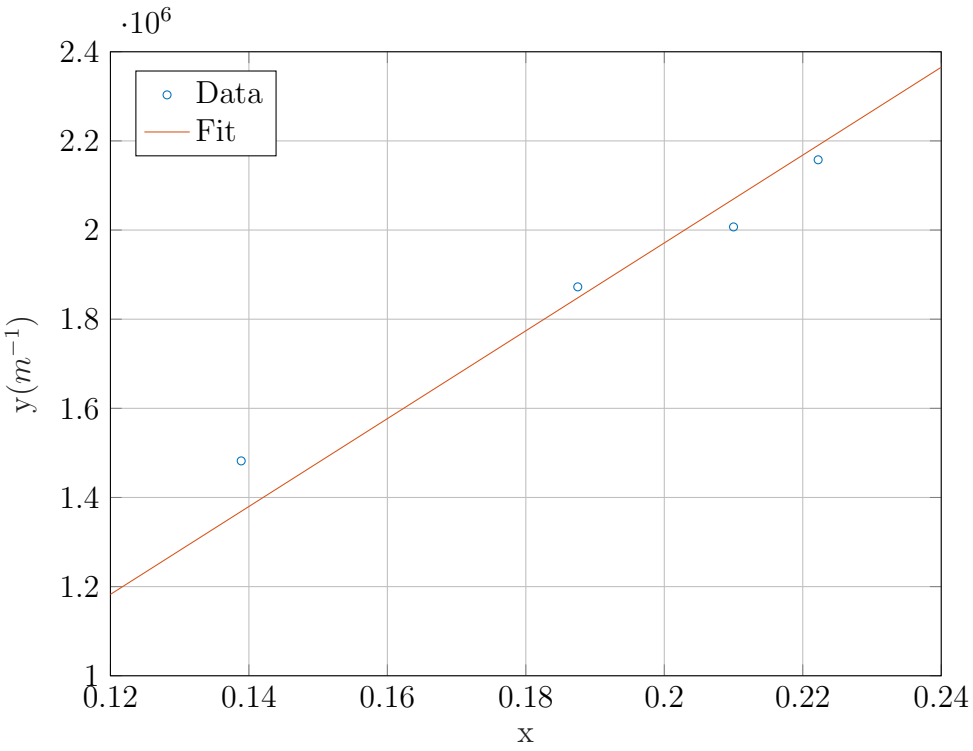


Figure 3: The slope is 9855295.02

If f is a function of x_is then the uncertainty in f is given by

$$\sigma_f^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \sigma_{x_i} \right)^2$$

The error in R is given by

$$\sigma_R = \lambda R(\sin^2\theta \cdot \sigma_d^2 + d^2 \cos^2\theta \cdot \sigma_\theta^2) = 1238.8$$

(Even though we get a different value for each wavelength, the difference is so little that it can be considered the same for all of them)

Result

The value of R_H is $98552.95 \pm 12.38\ cm^{-1}$.

The values for wavelength are given in the table below.

Colour	$\lambda(nm)$	$R(m^{-1})$
Violet	463.47	9.71E+06
Blue	498.25	9.56E+06
Green	534.03	9.99E+06
Red	674.74	1.07E+07

The values for the wavelengths for mercury spectrum were taken from:

`http://hyperphysics.phy-astr.gsu.edu`